



PLANNING ML PROJECTS

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Data Science as a Logistics Problem

- An undervalued perspective on ML projects is that they are directly analogous to supply chain problems.
- Raw materials for creating products $\leftrightarrow$ data
- Pipelines for getting results to users $\leftrightarrow$ Supply chain logistics

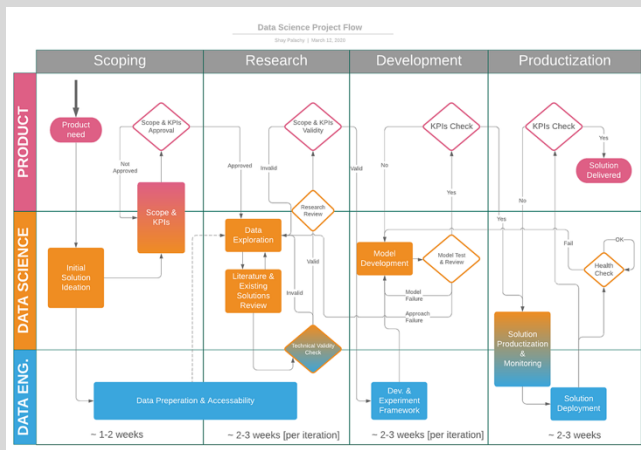
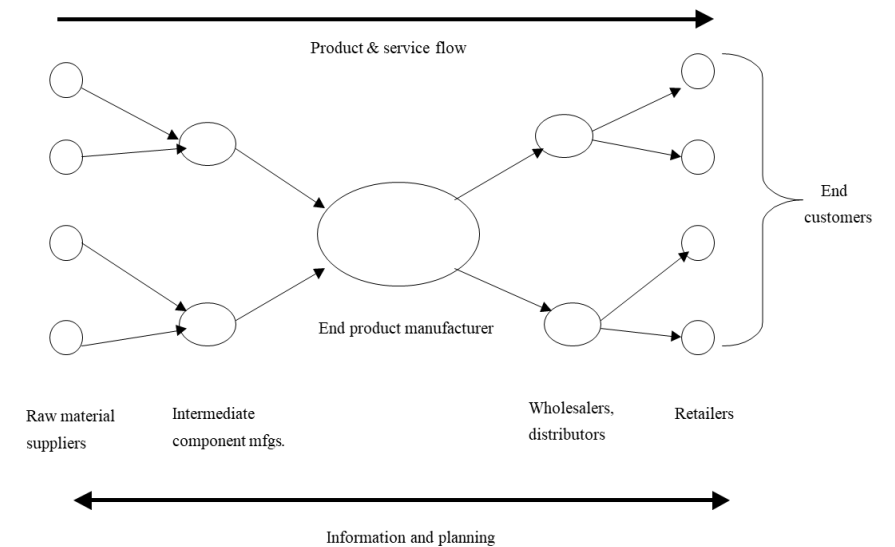


Figure 1.1 A Generic Supply Chain



Dependencies in the Plan

- Some steps can be done at the same time (parallel)
- Other steps cannot be started before previous steps are completed (series)
 - Can't build a model without data
- Some can be done after others are partially completed
- Can't do much until business needs is precisely articulated
- What data do you need
- Can you design a dashboard before modeling is completed?
- **Rate-Limiting:** a rate-limiting step is the step in the process that (1) has a bottleneck and (2) making it go faster has the most direct impact on the overall time

Rate-Limiting Factors

rate limiting factor is cooking stations, with the lowest rate

simple problem: clear, SEQUENTIAL pipeline

after improving cooking, new rate is 48 (checking), not 50.4 (improved cooking)

- Simple Question: During the dinner rush, orders require food preparation, cooking, plating, and final checking. The preparation station can support 50 meals per hour, the cooking stations 42 meals per hour, plating 60 meals per hour, and final checking 48 meals per hour. Determine the restaurant's maximum meal-completion rate. Management can increase the capacity of exactly one station by 20%. Which station should it improve if the goal is to increase total throughput?
- More complex: An instructor must prepare lecture notes, create slides, and write a quiz. Preparing the lecture notes takes 3 hours. Creating the slides takes 2 hours and cannot begin until the lecture notes are complete. Writing the quiz takes 4 hours and can be done independently of the other two tasks. If different people can work simultaneously, what tasks should begin first, and what is the minimum time needed to finish all three?



best solution: notes -> quiz -> slides, total time is 5 hours

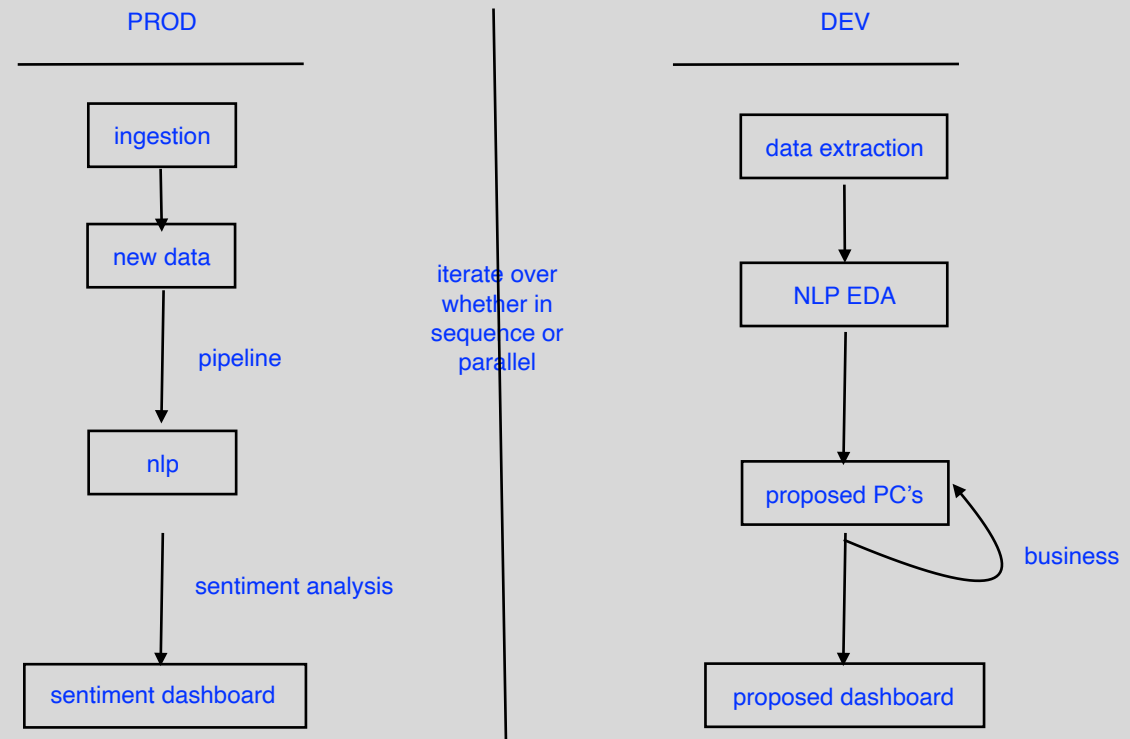
Basic Divisions of a ML Project

- EDA vs. Build
- Data vs. Model
- Dev vs Prod
- Questions to ask
 - What are the dependencies
 - What can be built in parallel
 - What needs to be kept
 - What are the resource constraints

Sketching Out Project Plans

- Returning to one of the 3 cases from ICA 1, let's see what we can do
 - Trying to get big picture—will return to smaller details next time

- Why do customers hate us?



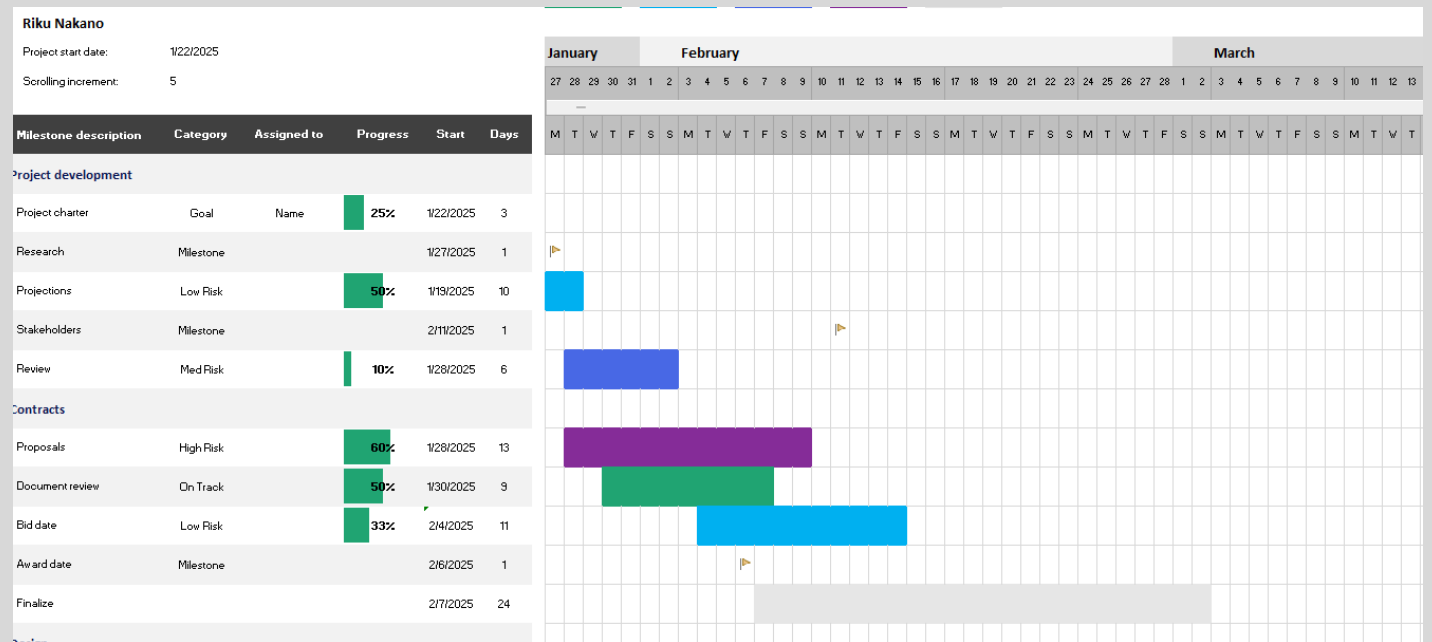
How to Plan—What to Ask

- Different projects will have different steps
 - Many project managers have a one-size-fits-all philosophy—don't let them bully you!
- We will focus on smaller projects and parts of larger projects
 - Larger projects have their own rules (and constraints)
- Projects always take longer than stakeholders think they should
 - Having a project plan allows you to explain why things take longer than they think

How to Plan—Planning Tools

- Gantt charts are very common tool for project planning
- What they are good for
 - Visualizing dependencies (series vs. parallel)
 - Accountability
 - Easily changeable (/s)

two kinds of Gantt charts:
1. ones that never change
2. ones that change daily



git in Data Science

- Git repositories are a basic tool for many data science organizations
- The more people that work on a project, the more essential git is for data science
- Why?
 - Version control
 - Track changes
 - Security settings
 - Imposes organization

Some terms in git

- **Repository** - "Database" of all the branches and commits of a single project
- **Branch** - Alternative state or line of development for a repository.
- **Merge** - Merging two (or more) branches into a single branch, single truth.
- **Main / Master** - Common name for the root branch, which is the central source of truth.
- **Stage** - Choosing which files will be part of the new commit
- **Commit** - A saved snapshot of staged changes made to the file(s) in the repository.
- **HEAD** - Shorthand for the current commit your local repository is currently on.
- **Push** - Pushing means sending your changes to the remote repository for everyone to see
- **Pull** - Pulling means getting everybody else's changes to your local repository
- **Pull Request** - Mechanism to review & approve your changes before merging to main/master

Git as a Planning Tool

- It can be valuable to have the planning tool where your project work is done
 - This is especially important when there are multiple collaborators on a project
- A big part of git is the documentation of accountability
 - Who
 - Changed what
 - When?
- Even in dev, lay out your git repository as if it were in production
 - A lot of bad habits come about because people can be sloppy in dev
 - More work to clean up than do right the first place